Advanced Dynare

Chapter IX

Method of Moments estimation of DSGE models

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Motivation

- Since Dynare 5, Method of Moments toolbox that provides functionality to estimate parameters by
  1. Simulated Method of Moments (SMM) up to any perturbation approximation order (with or without pruning)
  2. Generalized Method of Moments (GMM) up to 3rd-order pruned perturbation approximation
- Toolbox is inspired by replication codes accompanied to Andreasen et al. (2018), Born and Pfeifer (2014), and Mutschler (2018)
- Dynare 6: support for (Bayesian) impulse response function matching
- Inspired by Christiano, Eichenbaum, et al. (2005) and Christiano, Trabandt, et al. (2010)

Outline

1. Computing moments in Dynare
2. Method of Moments
   - SMM
   - GMM
   - IRF Matching
3. Dynare implementation

Dynare's stochastic simulation framework

- General DSGE model with shocks u_t, variables y_t, structural parameters θ, and perturbation parameter σ can be written as

$$0 = E_t f(y_{t+1}, y_t, y_{t-1}, u_t|\theta) \tag{1}$$

$$u_t = \sigma\eta_t, \quad \text{with} \quad \eta_t \overset{iid}{\sim} N(0,\Sigma) \tag{2}$$

- Note that

$$E[u_t] = 0 \tag{3}$$

$$E[u_t u_t'] = \Sigma_u = \sigma^2\Sigma \tag{4}$$

- Sidenote: shock distributions other than Gaussian are not (yet) supported by Dynare
- The solution, i.e. the policy functions take the generic form

$$y_t = g(x_{t-1}, u_t|\theta) \tag{5}$$

DSGE solution: perturbation

- Perturbation solution involves taking a Taylor-approximation around the non-stochastic steady-state (where y_t are all endogenous and x_t are states)
- Solution at arbitrary order then takes the form:

$$\begin{aligned} y_t =& \bar{y} + g_x(x_{t-1} - \bar{x}) + g_u u_t \\ &+ \frac{1}{2}[g_{xx}(x_{t-1} - \bar{x}) \otimes (x_{t-1} - \bar{x}) + 2g_{xu}(x_{t-1} \otimes u_t) + g_{uu}(u_t \otimes u_t) + g_{\sigma\sigma}\sigma^2] \\ &+ \frac{1}{6}[\dots] \\ &+ \dots \end{aligned} \tag{6}$$

How to compute moments?

- Want to pin down some of the parameters in θ by some form of moment matching
- Two ways of computing moments:
  1. Via simulation (`periods` option): Use approximated policy function to simulate data and compute the mean, covariance, auto-covariance, skewness, kurtosis, ...
  2. Use theoretical expressions based on state space representation
     $\rightarrow$ did that e.g. for the Kalman filter initialization

First-order approximation and moments

- Consider first order approximation of policy rule:

$$y_t = \bar{y} + g_x(x_{t-1} - \bar{x}) + g_u u_t, \tag{7}$$
$$x_t = \bar{x} + h_x(x_{t-1} - \bar{x}) + h_u u_t \tag{8}$$

- Unconditional first and second moments are

$$E[y_t] = \mu_y = \bar{y} \tag{9}$$
$$E[x_t] = \mu_x = \bar{x} \tag{10}$$
$$\Sigma_y(0) = E\left[(y_t - \bar{y})(y_t - \bar{y})'\right] = g_x \Sigma_x(0)\, g_x' + g_u \Sigma_u g_u' \tag{11}$$
$$\Sigma_x(0) = E\left[(x_t - \bar{x})(x_t - \bar{x})'\right] = h_x \Sigma_x(0)\, h_x' + h_u \Sigma_u h_u'\ , \tag{12}$$

where $\Sigma_x(0)$ is fixed point of Lyapunov equation

- Result allows computing auto-covariogram at lag j: $\Sigma_x(j)$ and $\Sigma_y(j)$

Higher-order approximation: the challenge

- Perturbation approximations are polynomials and thus have unavoidable oscillations
- Higher order perturbation policy functions do not inherit properties like monotonicity and convexity from the true policy functions
- Moreover, no way to control where such oscillations occur

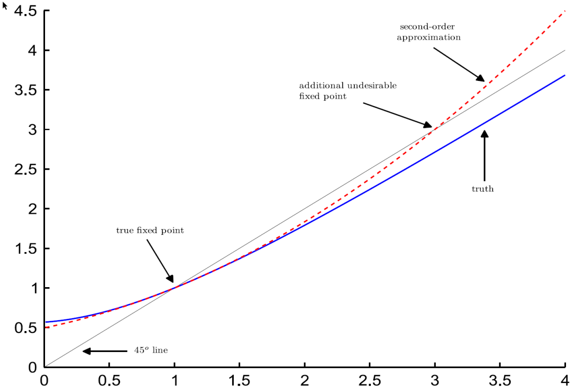
$\rightarrow$ could occur close to steady state

- Consider a univariate second-order example (setting uncertainty correction $\frac{1}{2}g_{\sigma\sigma}\sigma^2 = 0$) with policy function:

$$x_t = g_x x_{t-1} + g_{xx} x_{t-1}^2 + g_u u_t, \quad |g_x| < 1, \quad g_{xx} > 0 \tag{13}$$

- $|g_x| < 1$ ensures that first-order approximation is stable and unique.
- Second-order approximation has two fixed-points: $\bar{x} = 0$ and $\bar{x} = \frac{1-g_x}{g_{xx}}$
  $\Rightarrow$ Once the model passes the second fixed point it will explode

Example from Den Haan and De Wind (2012)



- Consider policy function with unique fixed point at 1 and globally stable dynamics

$$x_t = \frac{0.9}{e} + x_{t-1} + 0.9e^{-x_{t-1}} \qquad (14)$$

- Second-order approximation inherits
  - increasing in x_{t-1}
  - strictly convex x_{t-1}
  - local stability as slope at steady state is smaller than 1

- Such a polynomial must have second fixed point and globally unstable dynamics
  $\rightarrow$ explosive for large shocks

A slightly different take

- As first noted in Kim et al. (2008), higher order perturbation solutions tend to explode due to the accumulation of terms of increasing order
- For example, in second order approximated solution, quadratic term at time t will be raised to the power of two in the quadratic term at $t+1$
- Resulting in quartic term will become a term of order 8 at $t+2$ and so on

Solving the explosiveness of higher-order approximations

- Possibility of explosive behavior in higher-order approximations
  $\rightarrow$ Model may not be stationary or does not have an ergodic probability distribution

- Solution: **Pruning**
  - Leave out terms in solution that have higher-order effects than the approximation order
    $\rightarrow$ squaring second order term one period later results in term of fourth order
  - Kim et al. (2008) and Andreasen et al. (2018): pruned state space is stationary and ergodic (conditional on regularity conditions for the shocks)
  - Lombardo and Uhlig (2018) or Lan and Meyer-Gohde (2013) provide theoretical foundation for this seemingly ad-hoc procedure

Pruning for univariate example

- Decompose state vector into 1st- and 2nd-order effects

$$x_t = x_t^f + x_t^s = g_x x_{t-1}^f + g_x x_{t-1}^s + g_{xx}(x_{t-1}^f)^2 + 2g_{xx}x_{t-1}^f x_{t-1}^s + g_{xx}(x_{t-1}^s)^2 + g_u u_t \quad (15)$$

- Stable solution: Prune terms that contain $x_t^f x_t^s$ and $(x_t^s)^2$ to get law of motions:

$$x_t^f = g_x x_{t-1}^f + g_u u_t \quad (16)$$

$$x_t^s = g_x x_{t-1}^s + g_{xx}(x_{t-1}^f)^2 \quad (17)$$

where

$$\left(x_t^f\right)^2 = g_{xx}^2 \left(x_{t-1}^f\right)^2 + 2g_{xx}g_u x_{t-1}^f u_t + g_u^2 u_t^2 \quad (18)$$

Univariate example

- Pruned solution can be compactly rewritten as a stable state-space system

$$\underbrace{\begin{bmatrix} x_t^f \\ x_t^s \\ (x_t^f)^2 \end{bmatrix}}_{z_t} = \underbrace{\begin{bmatrix} g_x & 0 & 0 \\ 0 & g_x & g_{xx} \\ 0 & 0 & g_{xx}^2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_{t-1}^f \\ x_{t-1}^s \\ (x_{t-1}^f)^2 \end{bmatrix}}_{z_{t-1}} + \underbrace{\begin{bmatrix} g_u & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 2g_x g_u & g_u^2 \end{bmatrix}}_{B} \underbrace{\begin{bmatrix} u_t \\ x_{t-1}^f u_t \\ u_t^2 - \sigma_u^2 \end{bmatrix}}_{\xi_t} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ g_u^2 \sigma_u^2 \end{bmatrix}}_{c} \tag{19}$$

- Formulation assures that ξ_t is mean 0
- Note: Even if u_t is Gaussian, ξ_t is not!
  $\rightarrow$ Higher-order statistics (HOS) may contain additional information for estimation

Pruned state space system

- Given an extended state vector z_t and an extended vector of innovations ξ_t, the pruned perturbation solution of a DSGE model can be rewritten as a linear time-invariant state-space system for any approximation order (Andreasen et al. 2018):

$$z_t = c + A z_{t-1} + B\xi_t \tag{20}$$
$$y_t = \bar{y} + d + C z_{t-1} + D\xi_t \tag{21}$$

- Remember: usually non-Gaussian!

- Straightforward (but tedious) to compute moments (very similar to first-order)

A note on the state space

- As can be seen, we now need to keep track of all components of the state variables instead of the just their sum, i.e. the original state
  $\rightarrow$ pruning introduces additional state variables not part of the original state space
- nth order pruned solution does not deliver exact solution, even if the true policy function is an nth order polynomial (but raising approximation order alleviates this)
- Fortunately, it seems as if the accuracy problems occur mostly outside the ergodic set of many economic problems
- However: there is no guarantee that this transfers to other applications
- Needing to keep track of the individual terms also reintroduces curse of dimensionality

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Calibration

- Calibration: choose parameter values to match model moments to the data moments
  $\rightarrow$ usually one-to-one mapping
- Dates back to at least Kydland and Prescott (1982), who used it to evaluate the performance of the RBC model
- Original idea: choose parameter values to make the model consistent with growth observations, but judge model's performance to explain business cycles
  $\rightarrow$ don't use same data for calibration and evaluation

Moment matching: basic idea

- Calibration often too restrictive:
    - no straightforward one-to-one mapping
    - parameters not identified by growth observations
    - unclear which targets to match
    - efficiency concerns when not jointly setting parameters
    - How to quantify uncertainty?
- Solution: formulate formal estimation problem: try to minimize the distance between model moments and data moments
    1. Simulated Method of Moments: model moments are computed via simulations
    2. Generalized Method of Moments: model moments are computed theoretically
    3. IRF matching: model IRFs are matched to empirical IRFs

Simulated Method of Moments: notation

- m_t: vector of empirical observations with T observations on variables whose moments

$$\frac{1}{T}\sum_{t=1}^{T} m_t \tag{22}$$

  are of interest (e.g. $y_t, c_t, y_t y_{t-1}, y_t^3$)

- $m_i(\theta)$: model counterpart of m, computed on basis of $T' = \tau T$ observations of artificial data generated using parameter values θ
  $\rightarrow$ multiple τ of empirical data

- W: positive definite weighting matrix (if you have more moments than parameters)

Choice of the weighting matrix

- Any positive definite weighting matrix works
- But: there is only one optimal weighting matrix W^{opt} minimizing the asymptotic variance of the estimator, i.e. having the tightest bands: the inverse of the long-run covariance matrix of the moments
  $\rightarrow$ weights moments with their precision (similar to GLS)
- Problem: long-run variance is unknown (depends on true parameter vector) and needs to be estimated using plug-in estimator
  $\rightarrow$ usually a Newey and West (1987)-type estimator with a Bartlett kernel
- Popular approach: two stage/iterated GMM/SMM
  1. use non-optimal, typically diagonal weighting matrix in first stage
     $\rightarrow$ consistent but inefficient parameter estimate
  2. Use parameter estimate to recompute long-run variance of moments
     $\rightarrow$ asymptotically efficient

Simulated Method of Moments estimator

- Goal: minimize distance $G(\theta)$ between empirical and model simulated moments:

$$G(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t - \frac{1}{\tau T}\sum_{i}^{\tau T} m_i(\theta) \tag{23}$$

- SMM estimator $\hat{\theta}_{\mathsf{SMM}}$ is the value that minimizes

$$\hat{\theta}_{SMM} = \underset{\theta}{\text{argmin}} \left(G\left(\theta\right)\right)' W G\left(\theta\right) \tag{24}$$

- Dynare provides rich suite of numerical tools to solve this minimization problem

Statistical properties of SMM estimator

- Under regularity conditions in Duffie and Singleton (1993) and when using the optimal weighting matrix W^{opt}:

$$\sqrt{T}(\hat{\theta}_{\mathsf{SMM}} - \theta) \rightarrow N\left(0, (1+1/\tau)(D'W^{opt}D)^{-1}\right) \tag{25}$$

- W is computed using a Newey and West (1987)-type estimator with a Bartlett kernel
- Sandwich matrix

$$D = E\left[\frac{\partial m_i(\theta)}{\partial \theta}\right] \tag{26}$$

  - must be full rank (local identification!)
  - expectation is approximated by averaging over simulated τT data points
  - derivative is computed numerically

Generalized Method of Moments: notation

- m_t: as before, vector of empirical observations with T observations on variables whose moments
$$\frac{1}{T}\sum_{t=1}^{T} m_t \tag{27}$$
are of interest (e.g. $y_t, c_t, y_t y_{t-1}, y_t^3$)

- $E[m(\theta)]$: counterpart of m_t, whose elements are computed on basis of unconditional theoretical moments of the DSGE model using parameter values θ

- W: positive definite weighting matrix (if you have more moments than parameters)

Generalized Method of Moments estimator

- Moments distance is now given by :

$$G(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t - E[m(\theta)] \tag{28}$$

- GMM estimator $\hat{\theta}_{\text{GMM}}$ is the value that minimizes

$$\hat{\theta}_{GMM} = \underset{\theta}{\text{argmin}} \left(G\left(\theta\right)\right)' W G\left(\theta\right) \tag{29}$$

Statistical properties of GMM estimator

- Under regularity conditions in Hansen (1982):

$$\sqrt{T}(\hat{\theta}_{\mathsf{GMM}} - \theta) \rightarrow N\left(0, (D'W_{opt}D)^{-1}\right) \tag{30}$$

- Sandwich matrix

$$D = E\left[\frac{\partial m(\theta)}{\partial \theta}\right] \tag{31}$$

  - must be full rank (local identification!)
  - can be computed either analytically or numerically

Overidentification test

- Method of moments estimation requires at least as many target moments (n_m) as estimated parameters (n_θ)
  - Fewer moments: model is under-identified
  - exactly as many moments as parameters: just identified
  - more moments: model is overidentified
- Overidentification allows constructing a general specification test of the model (Hansen 1982):
- H_0: Overidentified model is correctly specified
- SMM:

$$T(1+1/\tau)(\hat{G}(\hat{\theta}_{SMM})'W^{opt}\hat{G}(\hat{\theta}_{SMM})) \rightarrow \chi^2(n_m - n_\theta) \tag{32}$$

- GMM:

$$T(\hat{G}(\hat{\theta}_{GMM})'W^{opt}\hat{G}(\hat{\theta}_{GMM})) \rightarrow \chi^2(n_m - n_\theta) \tag{33}$$

Penalized method ofmMoments

- Global minimum often found in regions of parameter space typically considered unlikely ("dilemma of absurd parameter estimates")
- Local minima often in more plausible regions and characterized by slightly worse values of the objective function
  $\Rightarrow$ include prior knowledge as an additional moment restriction (similar to adding priors to likelihood in Bayesian MCMC estimation)

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \begin{bmatrix} G(\theta) \\ \theta_{\text{prior}} - \theta \end{bmatrix}' \begin{bmatrix} W & 0 \\ 0 & \Omega_{\text{prior}}^{-1} \end{bmatrix} \begin{bmatrix} G(\theta) \\ \theta_{\text{prior}} - \theta \end{bmatrix} \tag{34}$$

- Caveat: leads to a loss in efficiency but may deliver (more) plausible results

Comparison to full-information methods

- Full-information (ML, Bayesian MCMC) methods are more efficient (consider all moments weighted with their precision), but suffer more from mis-specification (model is DGP)

- Limited-information methods like SMM/GMM are less efficient, but suffer less from mis-specification

- SMM/GMM can be more flexible: researchers often care not about most precisely estimated moments but select set
  $\rightarrow$ Method of Moments naturally allows putting more weight on those of interest

- Efficiency of Method of Moments depends on informativeness of moments (check out `dynare_sensitivity` and `identification`)

(Bayesian) IRF Matching

- Estimate key parameters by matching impulse response functions
- Idea: treat the empirical impulse responses as data and select parameters to ensure the model's impulse responses closely mirror their empirical counterparts
- Allows focusing on model with only shocks of interest instead of full DGP

(Bayesian) IRF Matching: Theory

- Assume you have estimated IRFs $\hat{\psi}$
- According to standard classical asymptotics:

$$\sqrt{T}\left(\hat{\psi} - \psi(\theta_0)\right) \overset{a}{\sim} \mathcal{N}(0,\ W(\theta_0, \zeta_0))\,. \tag{35}$$

  where
  - θ_0 are the *true* estimated parameters
  - ζ_0 are the *true* non-estimated parameters
- Treat $\hat{\psi}$ as "data", possibly combine with priors, and choose a value for θ_0 such that model IRFs $\psi(\theta)$ align as close as possible to $\hat{\psi}$

Practical concerns

- Dynare does not perform the SVAR or local projection estimation
- User has to provide:
  - the values of the empirical IRFs intended for matching
  - their importance by choosing an appropriate weighting matrix
- Standard practice in impulse response matching: diagonal weighting matrix with inverse of squared standard error of the empirical impulse response
- Correlations between components of different impulse response functions are also feasible, but often lack interpretation
- Often, we do transformations on empirical IRFs for reporting results (e.g. adding constants, cumsum, multiplying/dividing by factors or trend variables, etc)
  $\rightarrow$ `irf_matching_file` to match “empirical IRFs” and “model IRFs”
- Currently only supports `simulation_method = stoch_simul` to generate IRFs

Outline

Method of Moment Toolbox

- Interface:
  - `matched_moments` block to specify targeted moments
  - `matched_irfs` block to specify targeted impulse response functions
  - `method_of_moments` command to run estimation
- Common interface
  - `varobs` statement declaring observables in dataset
  - `estimated_params` block (and its derivatives) to specify parameters and penalties (priors)

varobs and matched_moments block

$$E(c_t)$$
$$E(y_t)$$
$$E(c_t^2)$$
$$E(c_t y_t)$$
$$E(c_t c_{t-1})$$
$$E(c_t c_{t-2})$$
$$E(c_t y_{t-3})$$

```
varobs y c;
matched_moments;
c;
y;
c*c;
c*y;
c*c(-1);
c*c(-2);
c*y(-3);
end;
```

- $E[c_t^2]$ is uncentered moment, while variance $\text{var}(c_t) = E[c_t^2] - E[c_t]^2$ is a centered moment
- By default, you declare UNCENTERED moments, i.e., $E[c_t^2]$, unless you use the `prefilter` option

varobs and matched_irfs block

rbc_irf_matching.mod

```
xx = [1.007,1.117,1.092];
ww = [51,52];

matched_irfs;
var log_y; varexo eps_g; periods 1, 2;  values 0.20, 0.17; weights 360, 140;
var ghat ; varexo eps_g; periods 2 3:5; values 1.01, (xx);  weights 50, 20;
var y    ; varexo eps_g; periods 10:11; values (log(1.05));  weights (ww);
end;
```

method_of_moments command: options

- Necessary options:

`method_of_moments(`	
`mom_method=GMM,`	possible values are `GMM`, `SMM`, or `irf_matching`
`datafile='MYDATA.mat')`	name of filename with data (unless IRF matching)

- Options common to GMM/SMM/IRF matching

`order=2,`	approximation order (GMM ≤ 3, any for SMM)
`penalized_estimator,`	use deviation from prior mean as additional moment and use prior precision as weight
`pruning,`	use pruning at orders>1; automatically triggered for GMM
`se_tolx=1e-6`	step size for numerical computation of std errors

`method_of_moments` command: options

- Options common to GMM/SMM

`weighting_matrix=`, `['DIAGONAL','OPTIMAL']`	weighting matrix (does iterated estimation) `IDENTITY_MATRIX`: use identity matrix `OPTIMAL`: optimal weighting matrix `DIAGONAL`: use diagonal of optimal weighting matrix
`weighting_matrix_scaling_factor=1`,	scaling of weighting matrix in objective

`method_of_moments` command: options

- Options only for SMM

`burnin=500`,	periods dropped at beginning of simulation
`bounded_shock_support`,	trim shocks in simulation to $\pm$ 2 stdev
`seed=24051986`,	seed used in simulations
`simulation_multiple=5`	multiple of data length used for simulation

- Options only for GMM

`analytic_standard_errors`,	compute standard errors using analytical derivatives

- Options only for IRF matching

`irf_matching_file`,	MATLAB file containing additional transformations on the model IRFs
`relative_irf`,	Requests the computation of normalized IRFs

Dynare output: GMM/SMM

RBC_MoM_GMM_order2.mod

Comparison of matched data moments and model moments (GMM)

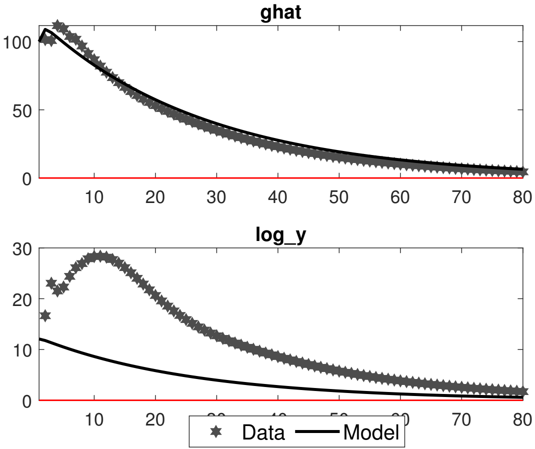
Moment	Data	Model
E[ln_c]	-0.1850290	-0.1852351
E[ln_n]	-0.9948910	-0.9947833
E[ln_iv]	-1.5650115	-1.5653406
E[ln_c*ln_c]	0.0351482	0.0352701
E[ln_c*ln_iv]	0.2914192	0.2918944
E[ln_n*ln_iv]	1.5563575	1.5564839
E[ln_iv*ln_iv]	2.4560278	2.4573826
E[ln_c*ln_n]	0.1836161	0.1837773
E[ln_n*ln_n]	0.9900779	0.9898778
E[ln_c*ln_c(-1)]	0.0351451	0.0352677

- Output is mostly similar to the one of `estimation`-command
- Results will be saved in `oo_.mom`
- Results of J-test will also be displayed in the Command Window:

```
Value of J-test statistic: 0.878744
p-value of J-test statistic: 0.999974
```

Dynare output: IRF matching

`rbc_irf_matching.mod`



- IRF matching will also display graphical comparison of empirical and model IRFs

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